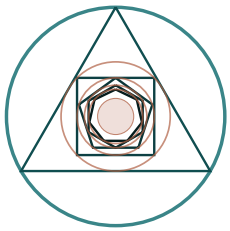
GEOMETRY • PROBABILITY • INFINITY

Almost Surely

The vanishing circle, a zillion triangles, and the limits of zero

Abstract. Begin with a circle, inscribe a regular polygon, place a circle inside it, and repeat forever. Increasing the polygon's number of sides can preserve a positive-radius core; repeating a fixed polygon collapses it. If the side count instead follows a random walk, the result changes sharply between a 50% and a 51% chance of gaining a side. Even a nonzero chance of a *zillion consecutive triangles* does not negate almost-sure survival: there is no smallest possible surviving circle, yet collapse has probability zero. If the upward probability itself approaches 50%, a second threshold appears. A comparison with a radiating black hole clarifies the difference between an unattained lower boundary and a zero limiting radius.



Illustrative nested polygons

What does "almost surely" mean? An event occurs *almost surely* if its probability is exactly 1. This is stronger than "extremely likely," but it does *not* mean every logically possible infinite sequence has that outcome. Exceptional sequences can exist and have probability zero. Thus, at 51% upward bias, collapse is possible along specially chosen infinite sequences, but it occurs with probability zero.

1 The geometry: an infinite product

Let $R_0 > 0$ be the initial circle's radius and let $N_k \geq 3$ be the number of sides of the kth inscribed regular polygon. Its incircle has radius

$$R_k = R_{k-1} \cos\left(\frac{\pi}{N_k}\right), \qquad R_m = R_0 \prod_{k=1}^{m} \cos\left(\frac{\pi}{N_k}\right). \tag{1}$$

Every finite R_m is positive, and R_m strictly decreases. Nevertheless, $R_\infty := \lim_{m\to\infty} R_m$ need not be zero. Indeed,

$$-\log\cos\left(\frac{\pi}{n}\right) = \frac{\pi^2}{2n^2} + O(n^{-4}), \qquad \boxed{R_\infty > 0 \iff \sum_{k=1}^{\infty} \frac{1}{N_k^2} < \infty}. \tag{2}$$

The equivalence follows by taking logarithms of (1); the asymptotic comparison is uniform for integer $n \geq 3$ up to fixed positive constants.

2 Two deterministic fates

A fixed polygon collapses. If $N_k = n$ at every step, then

$$R_m = R_0 \cos^m(\pi/n) \longrightarrow 0.$$

In particular, m successive triangles leave radius $R_0/2^m$. No finite step gives zero; the *limit* does.

Increasing side counts preserve a core. If $N_k = k + 2$, then the criterion (2) holds because $\sum_{n\geq 3} n^{-2} < \infty$. In fact,

$$R_\infty = R_0 \prod_{n=3}^{\infty} \cos\left(\frac{\pi}{n}\right) \approx \boxed{0.11494204485\, R_0}. \tag{3}$$

Infinitely many contractions preserve about 11.49% of the original radius. The difference is not whether shrinking stops; it never does. The difference is whether its cumulative effect converges.

3 The 50% versus 51% random walk

Start with $N_1 = 3$. Independently at each step, increase the number of sides by one with probability p, or decrease it by one with probability $q = 1 - p$. A downward move at three is reflected: it leaves another triangle. Thus,

$$N_{k+1} = \begin{cases} N_k + 1, & \text{probability } p, \\ \max(3, N_k - 1), & \text{probability } q. \end{cases} \tag{4}$$

For $p \le 1/2$, this reflected walk revisits three infinitely often almost surely. Each such visit halves the current radius, so $R_\infty = 0$ almost surely.

For $p > 1/2$, the strong law of large numbers gives $N_k/k \to 2p - 1$ almost surely. Consequently $\sum 1/N_k^2$ converges, so $R_\infty > 0$ almost surely.

Chance of gaining a side	50%	51%
Infinite limiting radius	Zero a.s.	Positive a.s.

4 A zillion triangles, with nonzero probability

Here is the counterintuitive objection. At $p = 0.51$, a downward toss has probability 0.49. Starting from the initial triangle, the probability that the first K nested polygons *are all triangles* is

$$\Pr(N_1 = \cdots = N_K = 3) = (0.49)^{K-1} > 0 \qquad \text{for every finite } K. \tag{5}$$

A trillion consecutive triangles have a nonzero probability. So does a *zillion* of them, if "zillion" means any specified finite integer. Such a run leaves

$$R_K = 2^{-K} R_0.$$

For any chosen threshold $\varepsilon > 0$, we can select a finite K so large that $2^{-K} R_0 < \varepsilon$. Therefore

$$\boxed{\Pr(0 < R_\infty < \varepsilon) > 0 \quad \text{for every } \varepsilon > 0, \qquad \Pr(R_\infty = 0) = 0 \quad (p = 0.51).} \tag{6}$$

These statements are not contradictory. The first quantifies over *each fixed finite length*; the second concerns an infinite limiting radius. The probability of triangles continuing forever from the outset is $\lim_{K\to\infty} 0.49^{K-1} = 0$.

In fact, at $p = 0.51$, the expected *total* number of visits to the triangular state, including the initial visit, is $1/(2p - 1) = 50$. There are almost surely only finitely many such visits. Arbitrarily long runs of *downward coin tosses* still occur somewhere in the infinite sequence, but most eventually happen while the polygon has too many sides for the streak to reach three.

5 There will always be a smaller circle

The permanent character of every inscription makes the following claim feel irresistible: if contraction never stops, the circle must ultimately vanish. But every contraction can be strict while the total multiplicative loss remains finite.

At 51% upward bias, equation (6) yields an even stranger statement. *Across possible complete infinite runs, there is no smallest positive surviving circle.* For any proposed positive radius ε, another possible run — beginning with sufficiently many triangles — has a limiting radius strictly between zero and ε, with positive probability. In probabilistic language,

$$\operatorname{ess\,inf} R_\infty = 0, \qquad R_\infty > 0 \quad \text{almost surely.}$$

There is a difference in the order of the words *every* and *one*: for every positive ε, some runs survive below it; it does **not** follow that one typical run lies below every positive ε. Each particular surviving run approaches its own fixed positive radius. The distribution of those radii has no positive floor.

Contrast an *actually infinite* nesting of triangles:

$$R_m = R_0\, 2^{-m} \longrightarrow 0.$$

Here zero is the actual limiting radius. At 51% bias, it is instead the unattained lower bound of possible positive survivor radii. These are distinct roles for zero, not distinct numbers.

6 Let the probability itself approach 50%

Now make the question evolve with the construction. At step k, set

$$\boxed{p_k = \frac{1}{2} + \frac{0.01}{k^\alpha}, \qquad q_k = 1 - p_k, \qquad \alpha > 0.} \tag{7}$$

The first upward probability is 51%. At every finite step it remains greater than 50%, but $p_k \downarrow 0.50$ as $k \to \infty$. Whether a central circle survives now depends on the *rate* at which the bias vanishes.

Bias above 50%	Rate	Limiting radius
$0.01/k^{1/4}$	Slow approach	Positive a.s.
$0.01/\sqrt{k}$	Critical approach	Zero a.s.
$0.01/k$	Fast approach	Zero a.s.

More generally, for the independent, reflected walk (4) with step-dependent probabilities (7),

$$\boxed{\Pr(R_\infty > 0) = \begin{cases} 1, & 0 < \alpha < 1/2, \\ 0, & \alpha \geq 1/2. \end{cases}} \tag{8}$$

Why $1/2$? Remove the reflection temporarily and let S_m be the sum of the independent ± 1 increments. Its accumulated expected drift is

$$\mathbb{E}S_m = 0.02 \sum_{k=1}^{m} k^{-\alpha} \sim \frac{0.02}{1-\alpha}\, m^{1-\alpha} \qquad (0 < \alpha < 1). \tag{9}$$

The centered fluctuations obey the law of the iterated logarithm and have magnitude on the order of $\sqrt{m \log\log m}$ along extreme subsequences. For $\alpha < 1/2$, the positive drift overwhelms those fluctuations: the reflected side count eventually grows like $m^{1-\alpha}$, making $\sum N_k^{-2}$ finite.

At $\alpha = 1/2$, the drift is only of order $\sqrt{m}$ and cannot overpower the negative excursions of order $\sqrt{m \log\log m}$. The same is true for $\alpha > 1/2$. The unreflected partial sums reach new negative minima infinitely often; reflection therefore forces infinitely many triangle visits, and the radius collapses.

7 Two approaches to zero

Approach I: infinitely many polygons at a fixed probability. At $p = 0.50$ the radius converges to zero. At any fixed $p > 0.50$, including $p = 0.51$, it converges to a *strictly positive* random limit almost surely.

Approach II: first take infinitely many polygons, then let $p \downarrow 0.50$. Every fixed $p > 0.50$ still gives a positive radius almost surely, yet the random limiting radius tends to zero in probability:

$$R_\infty(p) \xrightarrow[p \downarrow 1/2]{\Pr} 0, \qquad \lim_{p \downarrow 1/2} \Pr(R_\infty(p) > 0) = 1, \qquad \Pr(R_\infty(1/2) > 0) = 0. \tag{10}$$

The magnitude tends to zero, while the binary *survival probability* jumps at the critical boundary. Under the natural coupling that uses the same uniform random tosses for every p, the first convergence can even be taken almost surely.

The distinction is partly a matter of which limit is taken first. If we couple experiments for different p using the same uniform random numbers at each step, then $R_\infty(p) \downarrow 0$ almost surely as $p \downarrow 1/2$, even though $R_\infty(p) > 0$ almost surely at every fixed $p > 1/2$. Being positive at every point *before* a critical parameter does not require being positive *at* that parameter.

8 A shining black hole: a moving horizon, not the same zero

The image of a central circle that perpetually contracts suggests an evocative comparison with a black hole that slowly radiates. For a *stationary, nonrotating, fixed-mass* black hole, the Schwarzschild radius is fixed:

$$r_{\mathrm{S}} = \frac{2GM}{c^2}.$$

If the black hole loses mass through hypothetical Hawking evaporation, its characteristic gravitational-radius scale changes with the mass. In the slowly evolving approximation one may write

$$r_{\mathrm{S}}(t) = \frac{2G\,M(t)}{c^2}, \qquad M(t)\downarrow \Longrightarrow\ r_{\mathrm{S}}(t)\downarrow .$$

This is the *shining black hole*: the relevant radius need not remain fixed when the mass does not. But the familiar Schwarzschild formula strictly describes a stationary geometry. In a genuinely time-dependent spacetime, the location of a global event horizon is not generally determined by substituting an instantaneous mass into that formula. The final stages of evaporation require quantum-gravity physics that is not settled.

The geometric analogy is therefore deliberately limited. There are two distinct mathematical possibilities in our nesting:

- **No smallest surviving circle:** at 51% bias, for every $\varepsilon > 0$ some outcomes have $0 < R_\infty < \varepsilon$, while $R_\infty = 0$ has probability zero.
- **A genuine zero limit:** for triangles nested without end, $R_m = R_0/2^m \to 0$. Zero *is* the limiting radius, although every finite circle remains positive.

If a model of evaporation has $M(t) \to 0$, its radius scale would tend to zero; if $M(t)$ instead tends to a positive remnant, its radius scale would tend to a positive value. Neither outcome is established by the nested-polygon exercise, and neither should be confused with the event horizon's role as a causal boundary rather than a curvature singularity.

Conclusion. An infinity of irreversible inscriptions need not consume the last circle. At 51% upward bias, another smaller possible survivor always exists, yet almost every complete random history leaves a positive radius. In a triangle-only nesting, by contrast, the infinite product actually equals zero. Allow the bias itself to approach 50%, and the rate of approach decides which fate obtains. *There will always be a smaller circle* is not equivalent to *there will be no circle.*

Notes. Random-walk conclusions assume independent tosses, reflection at three sides, and exact real-number arithmetic; "almost surely" means probability one. For the physical comparison: NASA, *Imagine the Universe!* (Schwarzschild radius); NASA, *Shedding Light on Black Holes* (Hawking radiation).